Current Electricity

Chapter [3] Class - XII

Manish Joshi

D.A.V. Lohaghat

May 22, 2025

Limitations of Ohm's Law

- Ohm's law has been found valid over a large class of materials.

Limitations of Ohm's Law

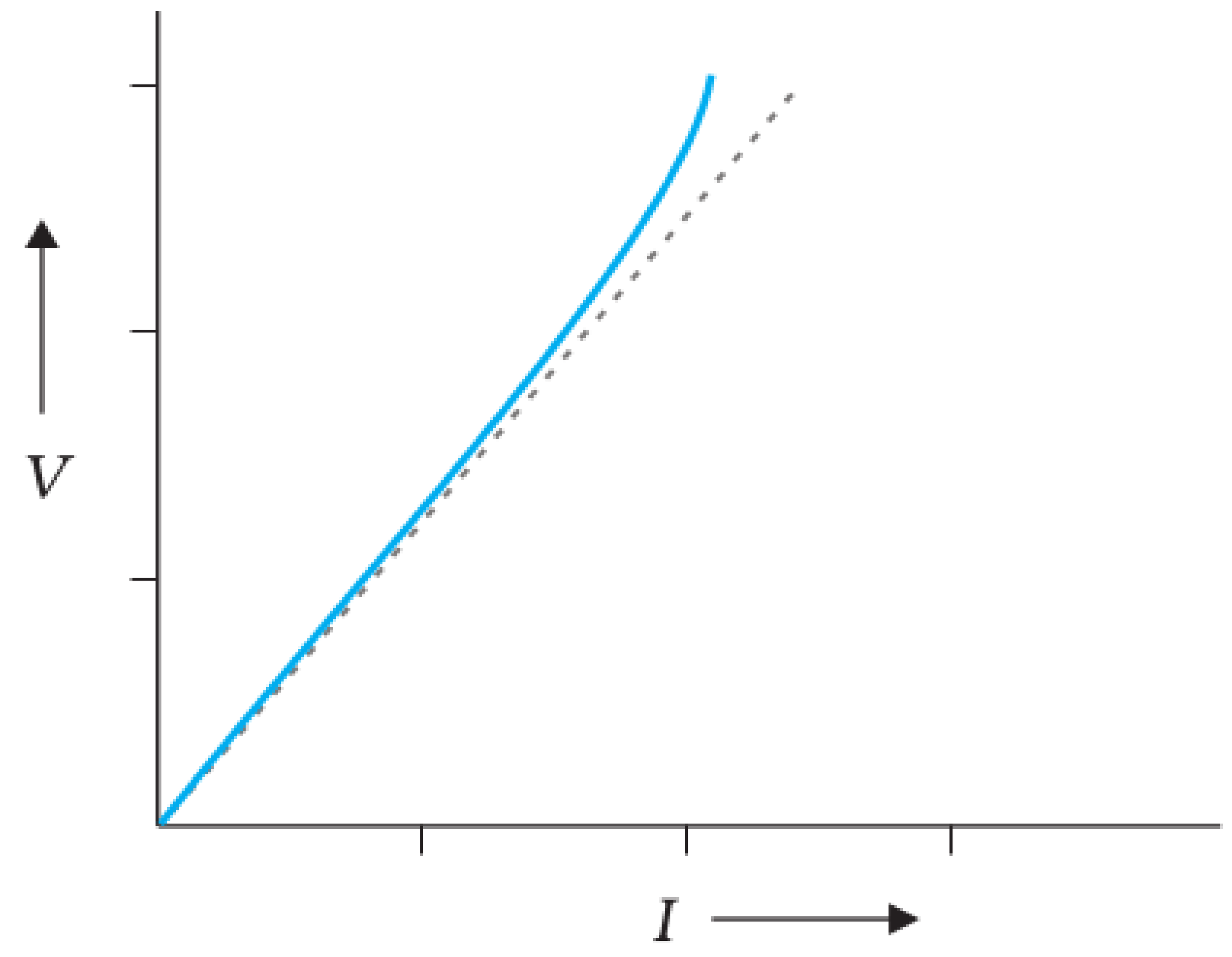


FIGURE 3.5 The dashed line represents the linear Ohm's law. The solid line is the voltage *V* versus current *I* for a good conductor.

- But, there do exist materials and devices in which Ohm's law doesn't hold (*i.e.*, $V \not\propto I$)
  a. V is not proportional to I for some conductor at high voltage.

Limitations of Ohm's Law

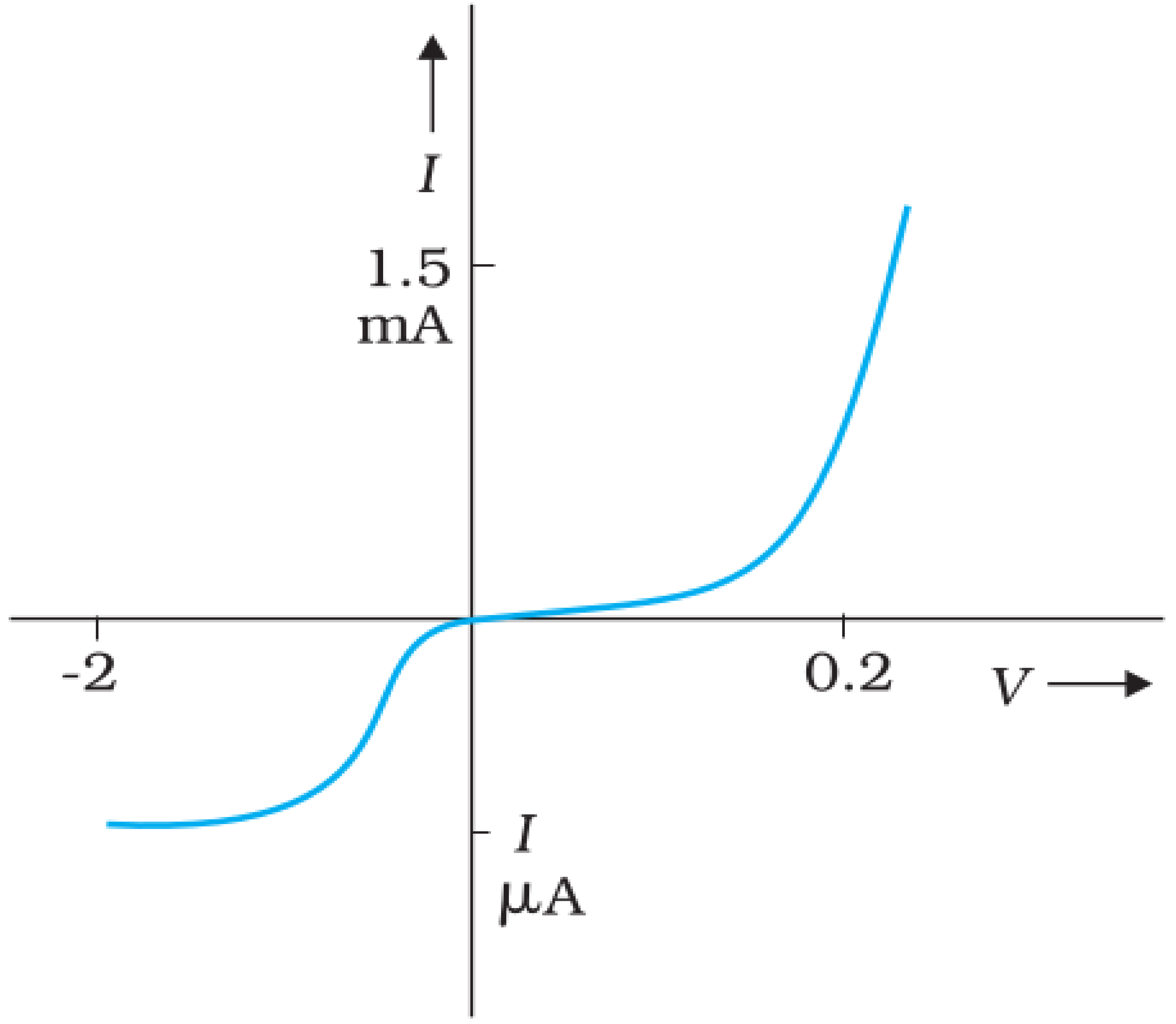


FIGURE 3.6 Characteristic curve of a diode. Note the different scales for negative and positive values of the voltage and current.

- But, there do exist materials and devices in which Ohm's law doesn't hold (*i.e.*, $V \propto I$)
  - (a) V is not proportional to I for some conductor at high voltage.
  - (b) For semicondouctors (diodes) the relation between V and I depends on the sign of V.

Limitations of Ohm's Law

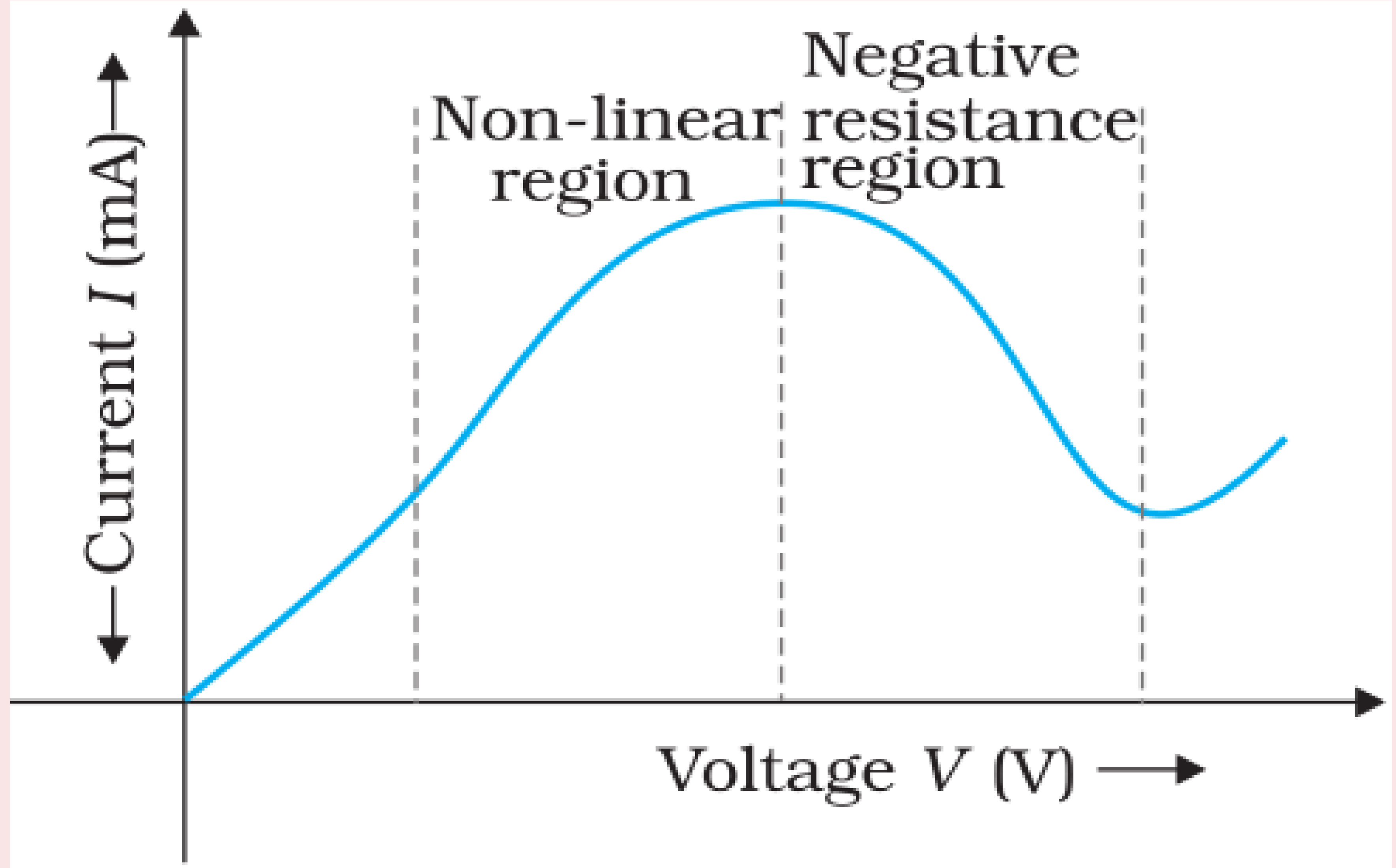


FIGURE 3.7 Variation of current versus voltage for GaAs.

- But, there do exist materials and devices in which Ohm's law doesn't hold (*i.e.*, $V \propto I$)
  - (a) V is not proportional to I for some conductor at high voltage.
  - (b) For semicondouctors (diodes) the relation between V and I depends on the sign of V.
  - (c) The relation between V and I is not unique, i.e., there is more than one value of V for the same current I . A material exhibiting such behaviour is GaAs.

RESISTIVITY OF VARIOUS MATERIALS

Material	Resistivity, ρ (Ω m) at 0°C
Conductors	
Silver	1.6×10^{-8}
Copper	1.7×10^{-8}
Aluminium	2.7×10^{-8}
Tungsten	5.6×10^{-8}
Iron	10×10^{-8}
Platinum	11×10^{-8}
Mercury	98×10^{-8}
Nichrome (alloy of Ni, Fe, Cr)	$\sim 100 \times 10^{-8}$
Manganin (alloy)	48×10^{-8}
Semiconductors	
Carbon (graphite)	3.5×10^{-5}
Germanium	0.46
Silicon	2300
Insulators	
Pure Water	2.5×10^{5}
Glass	$10^{10} - 10^{14}$
Hard Rubber	$10^{13} - 10^{16}$
NaCl	$\sim 10^{14}$
Fused Quartz	$\sim 10^{16}$

The materials are classified as **conductors**, **semiconductors** and **insulators** depending on their resistivities.

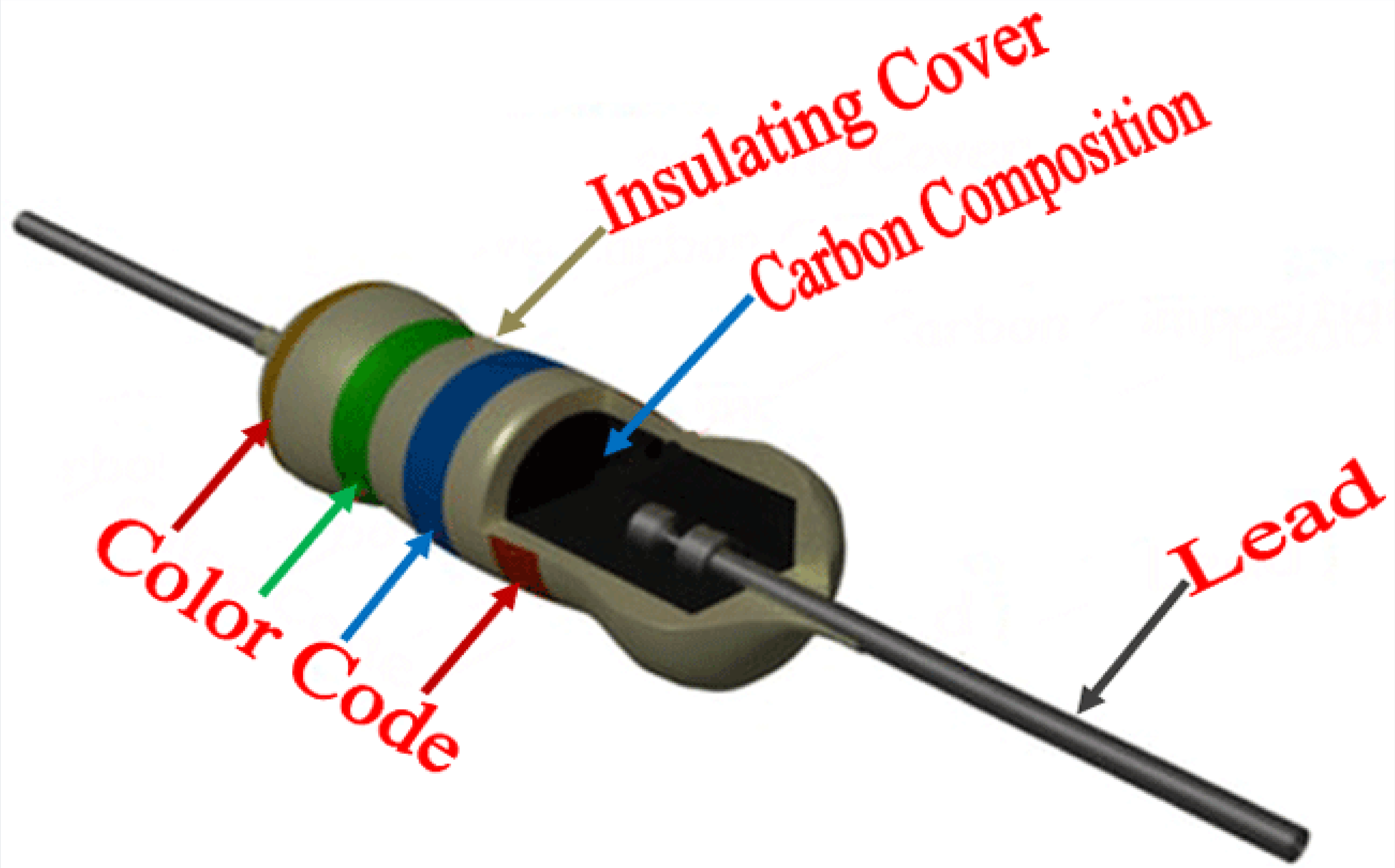
- **Conductors (Metals)** have **low resistivities** in the range of $10^{-8}\ \Omega m$ to $10^{-6}\ \Omega m$.
- **Insulators** like ceramic, rubber and plastics have **high resistivities**. $\approx 10^{18}$ times greater than metals or more.
- **semiconductors** (like Silicon, Germanium etc) have resistivities **in between** the **Conductors** and **Insulators**.

Types of Resistors

There are mainly **two types of Resistors** used in circuits and laboratories :

1. **Wire bound resistors** :
   - They are made by winding the wires of an alloy, viz., manganin, constantan, nichrome or similar ones.
   - They are **expensive**, **large**, but their **resistivity is insensitive to temperature.**
   - These resistances are typically in the **range** of a fraction of **an ohm to a few hundred ohms**.

Types of Resistors



There are mainly **two types of Resistors** used in circuits and laboratories :

1. **Wire bound resistors** :
2. **Carbon resistors** :
   - Made up of **Carbon compounds**.
   - Carbon resistors are **compact**, **inexpensive** and thus used in small electronic circuits.
   - Because of their **small size**, their **values** are given using a **colour code**.

Colour Codes for Carbon Resistors

- The first two rings give the first two significant figures of resistance in ohm.
- The third ring indicates the decimal multiplier.
- The last ring indicates the tolerance in per cent about the indicated value. Eg. $AB \times 10^C \pm D\%\ \Omega$
- Short Trick : **B B ROY of Great Britain has Very Good Wife**

Colour Codes for Carbon Resistors

TEMPERATURE DEPENDENCE OF RESISTIVITY

- The **resistivity** of a material is found to be **dependent on the temperature**.
- Different materials do not exhibit the same dependence on temperatures.
- Over a limited range of temperatures(ie., not too large), the **resistivity** of a **metallic conductor** is approximately given by,

$$\boxed{\rho_{\mathbf{T}} = \rho_0 \left[1 + \alpha(T - T_0)\right]} \tag{1}$$

where, $\rho_{\mathbf{T}} \implies$ **Resistivity at a temperature T**
$\rho_0 \implies$ **Resistivity at reference temperature $\mathbf{T_0}$**.
$\alpha \implies$ **Temperature coefficient of resistivity.**

TEMPERATURE DEPENDENCE OF RESISTIVITY

- Multiplying both sides by $\frac{l}{A}$ of eq(1), We get.

$$\rho_T \frac{l}{A} = \rho_0 \frac{l}{A} [1 + \alpha(T - T_0)]$$

$$\boxed{R_T = R_0 [1 + \alpha(T - T_0)]} \qquad \because R = \frac{\rho l}{A} \tag{2}$$

where, $\mathbf{R_T}$ $\implies$ **Resistance at a temperature T**
$\mathbf{R_0}$ $\implies$ **Resistance at reference temperature $\mathbf{T_0}$.**
α $\implies$ **Temperature coefficient of resistivity.**

TEMPERATURE DEPENDENCE OF RESISTIVITY

- The **dimensions** of α is $[Temperature^{-1}]$ or $[K^{-1}]$
- For **metals**, **Temp. Coefficient of resistivity** (α) is **positive.**
- The above **linear relation** between resistivity (ρ) and Temp. (T) eq(1) does **not hold for all material**.
- Some materials like **Nichrome** (which is an alloy of nickel, iron and chromium), **Manganin and constantan** (used in wire bound standard resistors) exhibit a very **weak dependence of resistivity with temperature**

TEMPERATURE DEPENDENCE OF RESISTIVITY

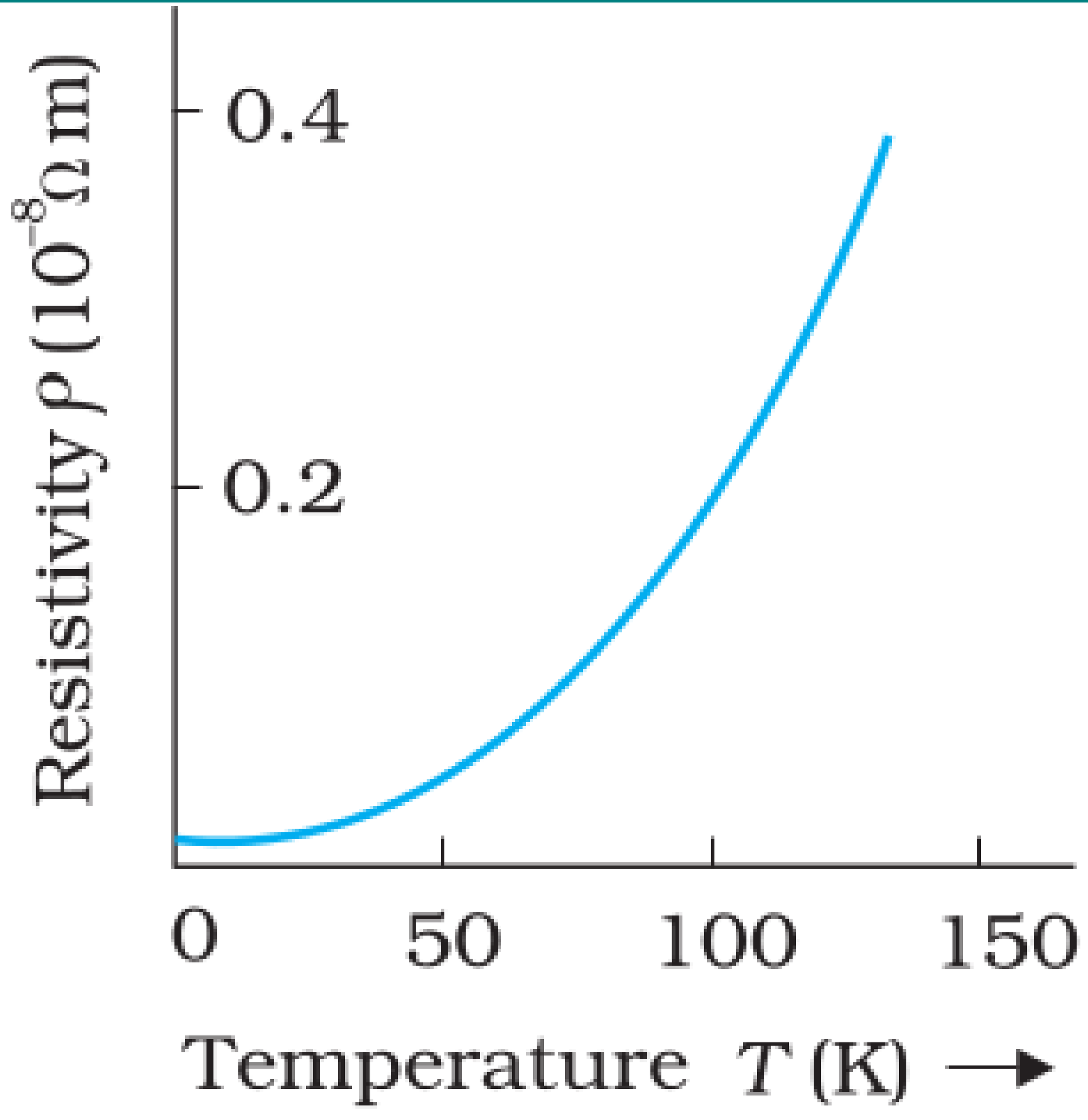


FIGURE 3.9 Resistivity ρ_T of copper as a function of temperature T.

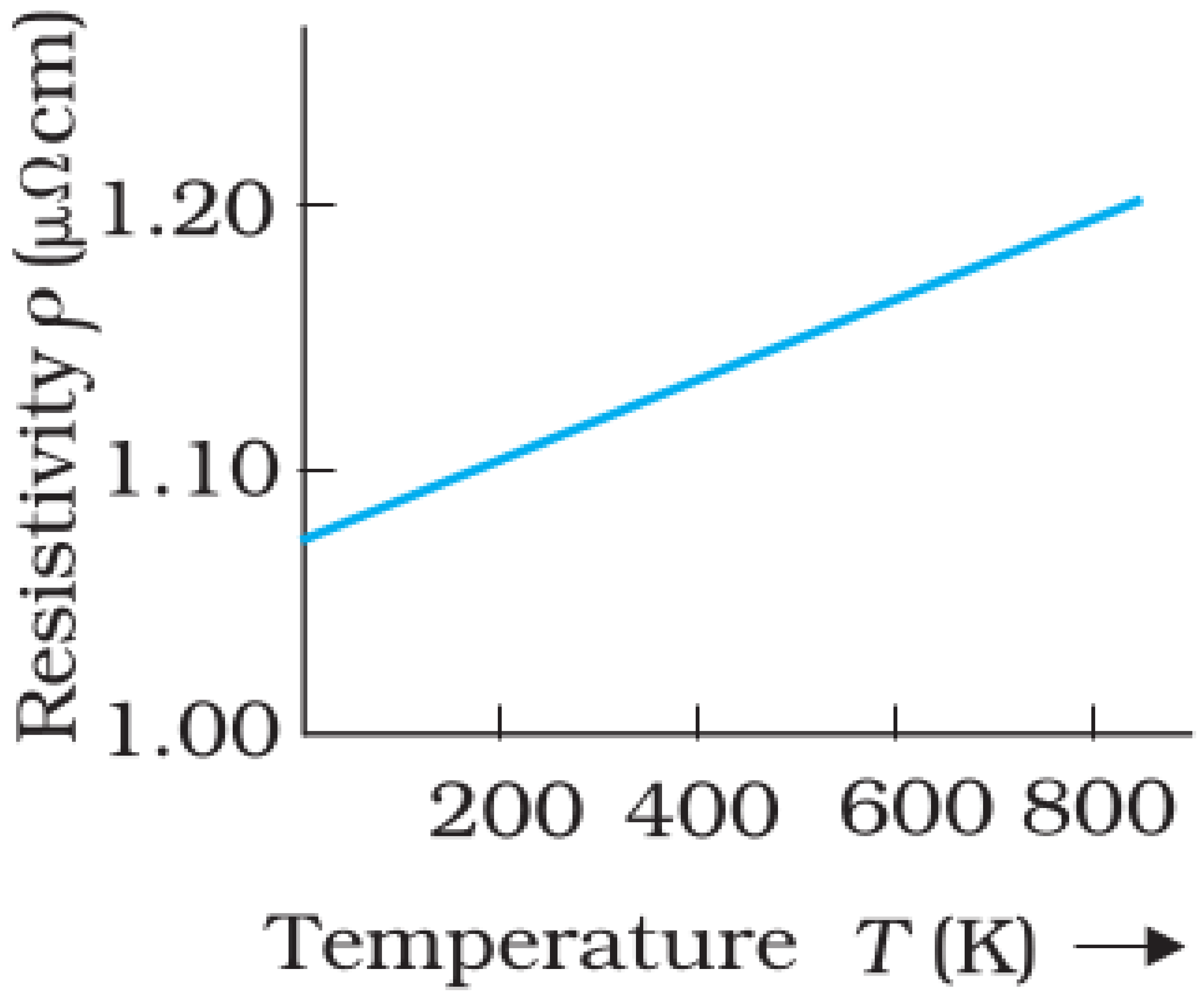


FIGURE 3.10 Resistivity ρ_T of nichrome as a function of absolute temperature T.

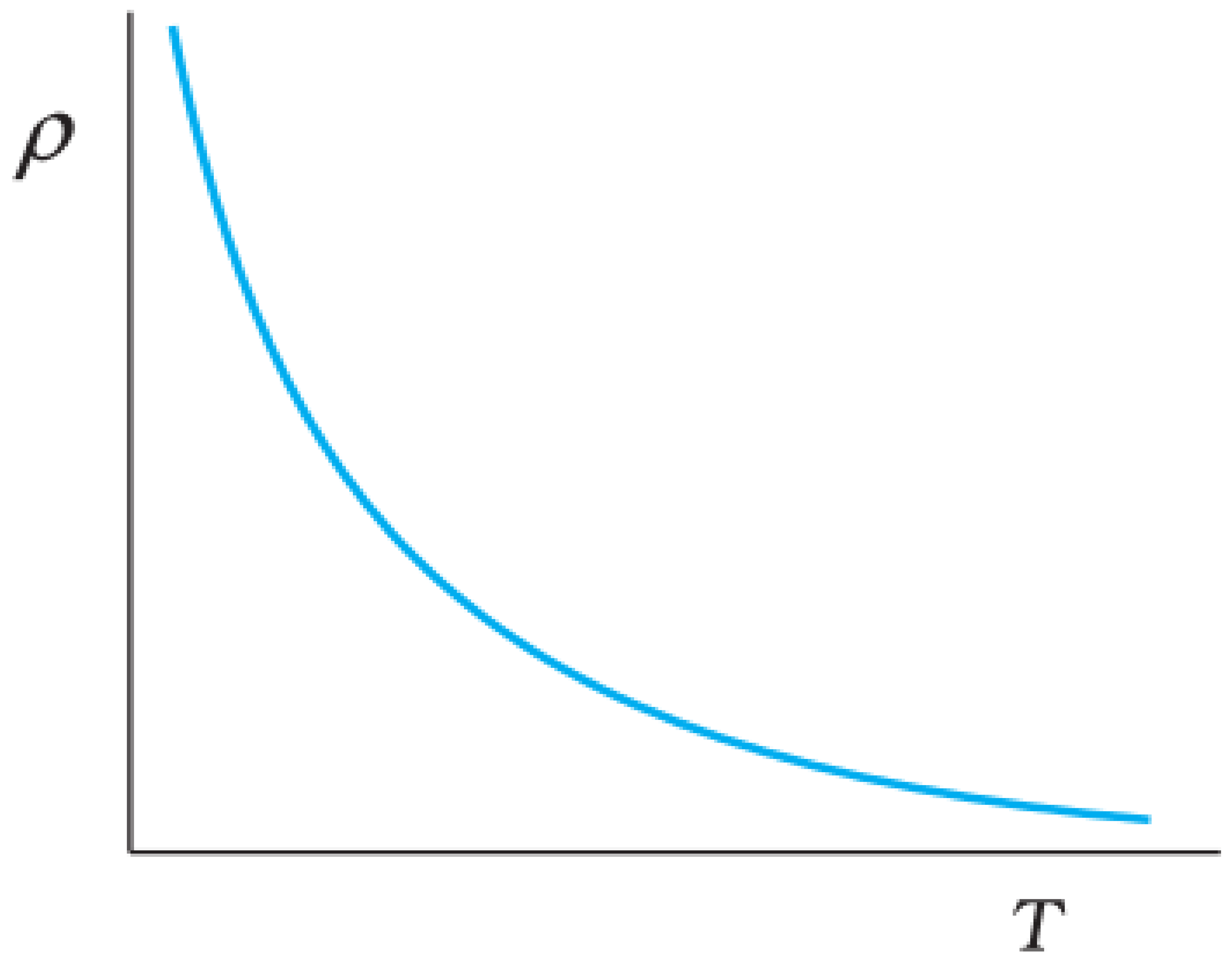


FIGURE 3.11 Temperature dependence of resistivity for a typical semiconductor.

Why resistivity ρ depends on temperature (T)

- **Resistivity** (ρ) of a material is given by-

$$\rho = \frac{m}{ne^2\tau} \tag{3}$$

- ρ thus **depends inversely** both on the **eletron no. density** n and on the **relaxation time** τ
- As we **increase temperature**, **average speed of the electrons increases** resulting in **more frequent collisions**. Thus τ **decreases with temperature.**
- **In a metal, resistivity increases with temperature.**
  Reason -relaxation time τ decrease with rise in temperature but n is **not dependent on temperature** to any appreciable extent.

Why resistivity ρ depends on temperature (T)

- **Resistivity** (ρ) of a material is given by-

$$\rho = \frac{m}{ne^2\tau} \tag{4}$$

- For **insulators and semiconductors resitivity** (ρ) **decreases with temperature**.
  Reason - number of conduction electrons per unit volume n increases with temperature. This increase more than compensates any **decrease in** τ.

Example 3.4

The resistance of the platinum wire of a platinum resistance thermometer at the ice point is 5 Ω and at steam point is 5.23 Ω. When the thermometer is inserted in a hot bath, the resistance of the platinum wire is 5.795Ω. Calculate the temperature of the bath.

Solution:

Given :

Resistance (R_0) at ice Point ($T_0 = 0^\circ C$) $= 5\ \Omega$

Resistance (R_{100}) at steam Point ($T_{100} = 100^\circ C$) $= 5.23\ \Omega$

Resistance (R_T) at hot bath (T) $= 5.795\ \Omega$

Example 3.4

Solution:

Given :

Resistance (R_0) at ice Point ($T_0 = 0^\circ C$) = 5 Ω

Resistance (R_{100}) at steam Point ($T_{100} = 100^\circ C$) = 5.23 Ω

Resistance (R_T) at hot bath (T) = 5.795 Ω

Step 1 : First we calculate the temp. of coefficient of resistivity α

$$\begin{aligned}
R_T &= R_0\left[1+\alpha(T-T_0)\right] \\
R_{100} &= R_0\left[1+\alpha(T_{100}-T_0)\right] \\
5.23\ \Omega &= 5\ \Omega\left[1+\alpha(100\ ^\circ C-0^\circ C)\right] \\
5.23 &= 5+500\alpha\ ^\circ C
\end{aligned}$$

Example 3.4

Solution:

$$\alpha = \frac{5.23-5}{500\ ^{\circ}C} = \frac{0.23}{500\ ^{\circ}C} = 0.00046\ ^{\circ}C^{-1}$$
$$\alpha = 4.6 \times 10^{-4}\ ^{\circ}C^{-1} \tag{5}$$

Step 2: Now, using this value of α claculating the temp. of the bath.

$$\begin{aligned} R_T &= R_0\left[1+\alpha(T-T_0)\right] \\ 5.795\ \Omega &= 5\ \Omega\left[1+4.6\times 10^{-4}\ ^{\circ}C^{-1}(T-0^{\circ}C)\right] \\ \frac{5.795}{5} &= 1+4.6\times 10^{-4}\ ^{\circ}C^{-1}(T) \end{aligned}$$

Example 3.4

Solution:

$$
\begin{aligned}
1.159 &= 1 + 4.6 \times 10^{-4}\ {}^\circ C^{-1}(T) \\
1.159 - 1 &= 4.6 \times 10^{-4}\ {}^\circ C^{-1}(T) \\
\frac{0.159}{4.6 \times 10^{-4}\ {}^\circ C^{-1}} &= (T) \\
T &= 0.03456 \times 10^{4}\ {}^\circ C \\
T &= 345.6\ {}^\circ C \qquad (6)
\end{aligned}
$$

14. Pieces of copper and of silicon are initially at room temperature. Both are heated to temperature T. The conductivity of **1**

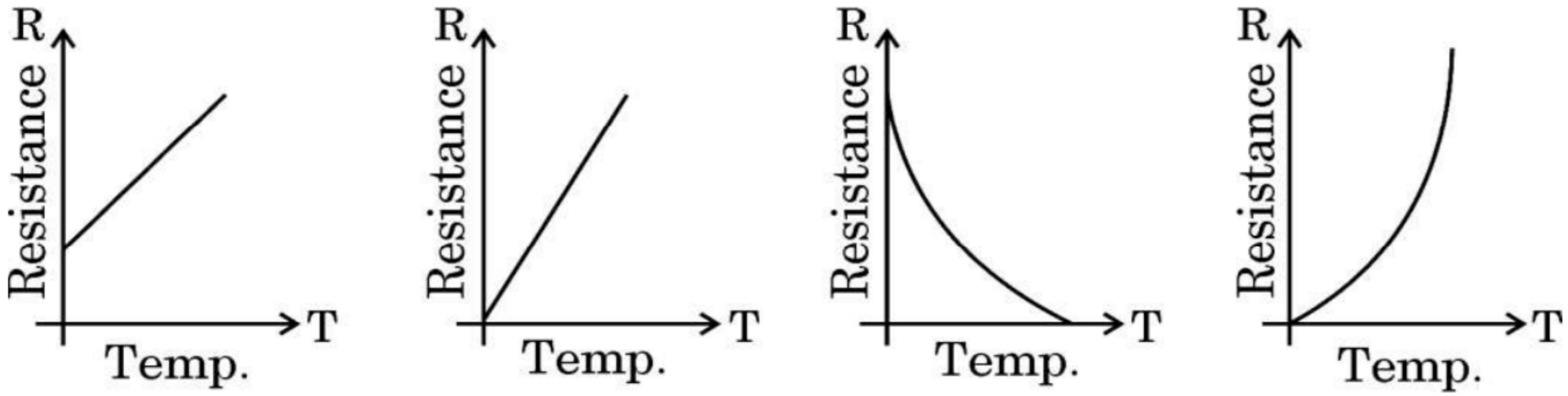
(a) both increases.

(b) both decreases.

(c) copper increases and silicon decreases.

(d) copper decreases and silicon increases.

2. For a metallic conductor, the correct representation of variation of resistance R with temperature T is :



(a) (b) (c) (d)

16. *Assertion (A) :* The temperature coefficient of resistance is positive for metals and negative for semi-conductors.

Reason (R) : The charge carriers in metals are negatively charged whereas in semiconductors they are positively charged.

24. (i) Define 'temperature coefficient of resistance' of a metal.

(ii) Show the variation of resistivity of copper with rise in temperature.

(iii) The resistance of a wire is 10 Ω at 27 °C. Find its resistance at –73 °C. The temperature coefficient of resistance of the material of the wire is $1.70 \times 10^{-4}\ °C^{-1}$.

$T_0 = 27°C$ $\quad$ $T = -73°C$ $\quad$ $R = ?$

$R_0 = 10\,\Omega$ $\quad$ $\alpha = 1.7 \times 10^{-4}\ °C^{-1}$

$R = R_0[1 + \alpha(T - T_0)]$

$R = 10[1 + 1.7 \times 10^{-4} \times (-73 - 27)]$

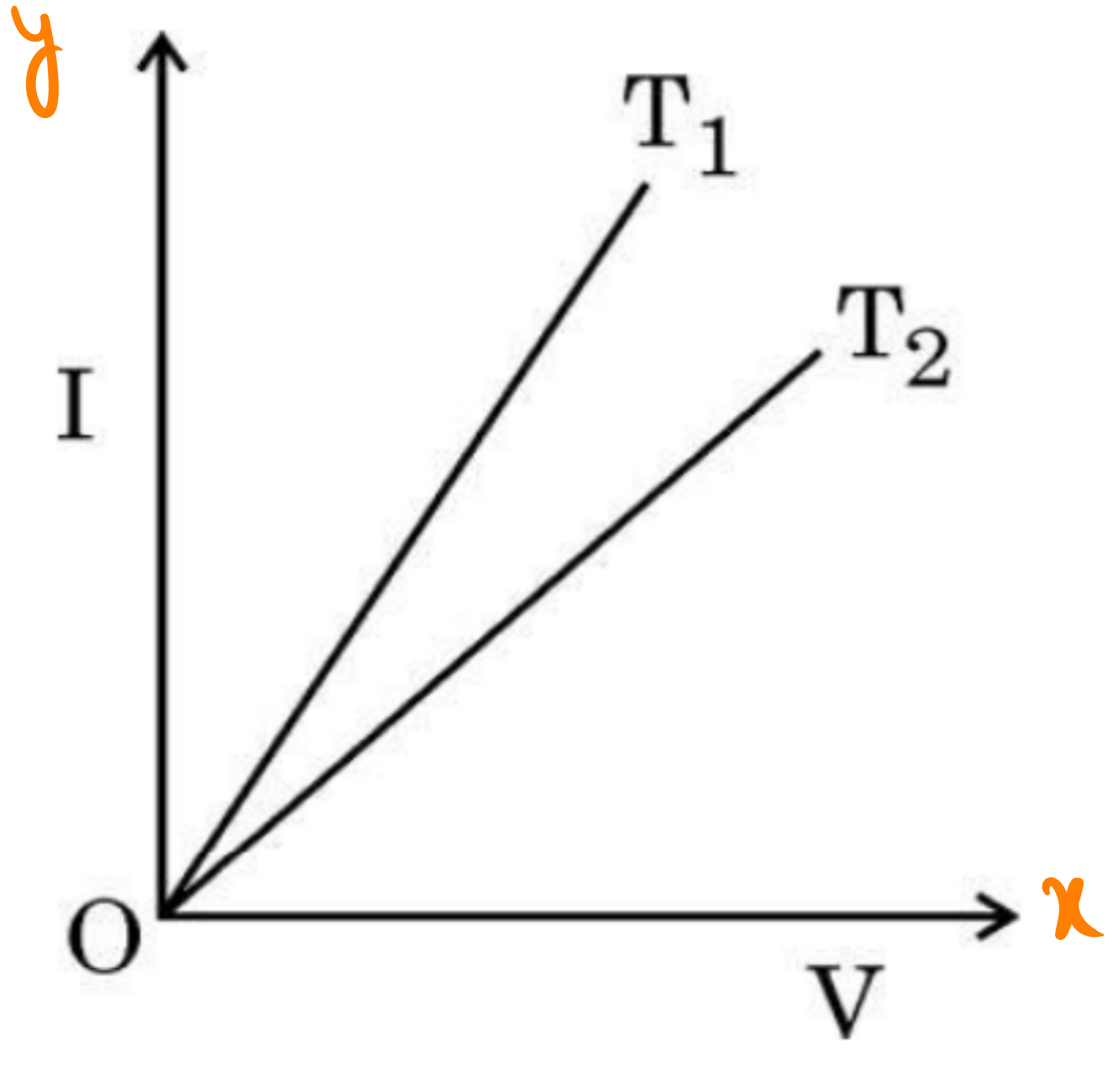
$R = 10[1 + 1.7 \times 10^{-4} \times (-100)$

$R = 10[1 - 0.017]\ \Omega$

$R = 10 \times 0.983$

$R = 9.83\,\Omega$

1. The figure shows the voltage (V) versus the current (I) graphs for a wire at two temperatures T_1 and T_2. One can conclude that :



$I = \frac{1}{R} V$

↳ slope

$\text{Slope1} > \text{Slope2}$

$\frac{1}{R_1} > \frac{1}{R_2}$

$R_2 > R_1$

$\Rightarrow T_2 > T_1$

(A) $T_2 = 2T_1$

(B) $T_1 > T_2$

(C) $T_1 = T_2 / 3$

(D) $T_1 < T_2$

2. The resistance of a wire at 20 °C is 50 Ω. When it is heated to 120 °C, the resistance becomes 51 Ω. The temperature coefficient of resistance of the material of the wire is **1**

(A) $2 \times 10^{-4}\ {}^\circ C^{-1}$ (B) $1 \times 10^{-4}\ {}^\circ C^{-1}$

(C) $5 \times 10^{-4}\ {}^\circ C^{-1}$ (D) $1 \times 10^{-3}\ {}^\circ C^{-1}$

$R_0 = 50\Omega$ $T_0 = 20^\circ c$ $\alpha = ?$

$R = 51$ $T = 120^\circ c$

$R = R_0[1 + \alpha(T - T_0)]$

$51 = 50[1 + \alpha[120 - 20)]$

$51 = 50[1 + 100\alpha]$

$51 = 50 + 5000\alpha$

$1 = 5000\alpha$

$\alpha = \frac{1}{5000} = 2 \times 10^{-4}\ {}^\circ c^{-1}$

21. The resistance of a wire at 25°C is 10·0 Ω. When heated to 125°C, its resistance becomes 10·5 Ω. Find (i) the temperature coefficient of resistance of the wire, and (ii) the resistance of the wire at 425°C. 2

$R_0 = 10\Omega$ $\quad T_0 = 25^\circ C$

$R = 10.5\Omega$ $\quad T = 125^\circ C$

$\alpha = ?$ $\quad R_{T'} = ?$ $\quad T' = 425^\circ C$

$R = R_0(1 + \alpha(T - T_0)]$

$10.5 = 10(1 + \alpha 100)$

$10.5 = 10 + 10^3\alpha$

$0.5 = 10^3\alpha$

$\alpha = \frac{0.5}{10^3} = 5 \times 10^{-4}\ {}^\circ C^{-1}$

$R' = R_0(1 + \alpha(T' - T_0)]$

$R' = 10(1 + 5\times10^{-4}\times400)$

$R' = 10(1 + 0.2) = 12\Omega$

20. Find the temperature at which the resistance of a conductor increases by 25% of its value at 27 °C. The temperature coefficient of resistance of the conductor is 2.0×10^{-4} °C^{-1}. **2**

$T_0 = 27°C$ $\quad R_0 \quad R = \frac{125}{100} R_0 = \frac{5}{4} R_0 \quad \alpha = 2 \times 10^{-4}$ °C^{-1}

$$R = R_0(1 + \alpha(T - T_0)]$$

$$\frac{5}{4}\cancel{R_0} = \cancel{R_0}(1 + 2\times10^{-4}(T-27)]$$

$$\frac{1}{4} = 2\times10^{-4}(T-27)$$

$$\frac{10^4}{8} = T - 27$$

$$T = \frac{10000}{8} + 27$$

$$= 1250 + 27$$

$$= 1277°C$$

17. Find the temperature at which the resistance of a wire made of silver will be twice its resistance at 20°C. Take 20°C as the reference temperature and temperature coefficient of resistance of silver at 20°C = $4{\cdot}0 \times 10^{-3}\ K^{-1}$. *2*

$T_0 = 20^\circ C$ R_0

T $R = 2R_0$ $\alpha = 4 \times 10^{-3}\ K^{-1}$

$R = R_0[1 + \alpha(T - T_0)]$

$2\cancel{R_0} = \cancel{R_0}[1 + 4\times10^{-3}(T-20)]$

$1 = 4\times10^{-3}(T-20)$

$\frac{1000}{4} = T - 20$

$250 + 20 = T$

$270 = T$

$T = 270^\circ C$

(ii) The temperature coefficient of resistance of nichrome is $1{\cdot}70 \times 10^{-4}\ {}^\circ C^{-1}$. In order to increase resistance of a nichrome wire by $8{\cdot}5\%$, the temperature of the wire should be increased by : *1*

(A) 250°C

(B) 500°C

(C) 850°C

(D) 1000°C

$R_0 \quad T_0$

$R = \frac{108.5}{100} R_0 \quad T$

$\alpha = 1.7 \times 10^{-4}\ {}^\circ C^{-1}$

$\frac{108.5}{100} \cancel{R_0} = \cancel{R_0}\left(1 + 1.7 \times 10^{-4} \Delta T\right)$

$\frac{8.5}{100} = 1.7 \times 10^{-4} \Delta T$

$\frac{8.5}{1.7 \times 10^{-2}} = 5 \times 10^{2}\ {}^\circ C = 500^\circ C$

(i) (a) Consider the contribution of the following two factors I and II in resistivity of a metal :

I. Relaxation time of electrons

II. Number of electrons per unit volume

The resistivity of a metal increases with increase in its temperature because :

(A) I decreases and II increases.

(B) I increases and II is almost constant.

(C) Both I and II increase.

(D) I decreases and II is almost constant.

OR